

[Figure 11-1](#) may remind you of sudoku, since it involves checking for the uniqueness of symbols in rows and columns. In sudoku, the goal is to fill in the numbers 1 through 9 such that each row, column, and 3×3 block contains exactly one instance of each number ([Figure 11-2](#)). Sudoku first gained popularity in Japan, and indeed a sudoku puzzle is reminiscent of the Japanese magic squares we explored in Chapter 2.

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			
				8			7	9

[Figure 11-2](#): An uncompleted sudoku grid (source: Wikimedia Commons)

It's an interesting exercise to think about how to write an algorithm that could solve sudoku puzzles. The simplest, slowest possible algorithm would rely on brute force: just try every possible combination of numbers and repeatedly check whether they constitute a correct solution, repeating until the solution is found. This would work, but it lacks elegance, and it could take an extremely long time. It doesn't seem intuitively right that filling in 81 numbers in a grid according to rules that anyone could easily follow